

Cybersecurity Threats in AI-Driven Construction Systems: A Framework for Adversarial Machine Learning Risks in the Built Environment

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Abstract

The rapid integration of Artificial Intelligence (AI) and Machine Learning (ML) technologies into construction workflows, including autonomous robotic systems, computer-vision-based quality inspection, AI-driven BIM platforms, and predictive maintenance engines, introduces a class of cybersecurity threats that existing construction-specific frameworks do not address. Prior frameworks, such as those grounded in the CIA triad or Parkerian hexad, were designed for conventional Information Technology (IT) and Operational Technology (OT) environments and fail to account for the unique attack surface presented by trained ML models deployed in safety-critical construction contexts. This paper identifies this gap and proposes a novel framework: the Adversarial Construction Intelligence Security (ACIS) framework. The framework characterises five categories of AI-specific threats, namely training data poisoning, adversarial input attacks, model inversion, model extraction, and supply chain model compromise, and maps them across four categories of construction AI assets: design intelligence systems, site perception systems, autonomous execution systems, and facility management AI. Drawing on first principles of adversarial machine learning, construction process theory, and systems engineering, we develop a risk matrix, threat taxonomy, and lifecycle security process tailored to construction environments. Construction-specific challenges are identified including the federated and multi-party nature of AI model training across project stakeholders, the physical consequences of adversarial inputs to autonomous machinery, and the intellectual property exposure of proprietary AI models embedded in digital twins. The ACIS framework complements and extends the prior systemic cybersecurity frameworks for construction, and provides researchers, BIM managers, project managers, and policymakers with a structured basis for understanding and mitigating AI-specific cyber threats in the built environment.

1. Introduction

1.1. AI Turn in Construction

The construction industry, historically among the slowest sectors to adopt digital technology [1,2], is undergoing a profound transformation. Turk et al. [3] characterised as Construction 4.0 the application of cyber-physical systems, IoT, and digital twins to the built environment is now evolving into an AI-intensive paradigm. Convolutional neural networks are deployed on construction sites to detect personal protective equipment (PPE) non-compliance [4]. Generative design algorithms propose structurally optimised building geometries [5]. Autonomous reinforced concrete robots execute rebar placement tasks guided by real-time vision models [6]. Digital twin platforms ingest sensor streams and use predictive AI models to schedule maintenance and predict structural fatigue [7,8].

This shift does not merely accelerate digitalization; it fundamentally changes the nature of the computational artefacts that mediate between human intent and physical construction outcomes. A trained neural network is not a deterministic piece of software that can be audited line by line. It encodes complex statistical patterns extracted from data, and its behaviour

is governed by those patterns in ways that can be manipulated by adversaries in manners entirely unlike conventional cyberattacks on IT or OT systems [9,10].

Despite this, existing cybersecurity frameworks for construction, including the seminal systemic framework of Turk et al. [3], do not address AI or ML systems as distinct assets requiring distinct threat models. General cybersecurity literature has developed the field of adversarial machine learning (AML) [11], which studies how ML models can be attacked through their inputs, training processes, and model artefacts. Yet this body of knowledge has not been brought to bear on the construction sector. This paper addresses that gap.

1.2. Definition of Adversarial Machine Learning in Construction Context

Adversarial machine learning refers to a body of attacks and defences focused on vulnerabilities inherent to the structure of trained ML models. Unlike conventional cyberattacks, which target data in transit or software logic, AML attacks target the statistical decision boundaries of models. A landmark taxonomy by Papernot et al. [12] distinguishes attacks by their goal (integrity violation, availability violation, confidentiality violation), the attacker's knowledge (white-box, black-box, grey-box), and the attack stage (training time vs. inference time). In construction contexts, these distinctions map onto specific, high-consequence scenarios.

For example, a white-box training-time poisoning attack on an AI-based structural defect detection model could cause the model to systematically misclassify a specific type of concrete crack with potentially catastrophic downstream consequences for structural safety. A black-box inference-time adversarial input attack on a site safety vision system could cause it to fail to detect an unsafe worker condition. A model extraction attack on a proprietary AI engine embedded in a digital twin could exfiltrate millions of dollars of intellectual property [13,14].

1.3. Security Breaches Involving AI Systems: Reported Cases

While AI-specific attacks on construction systems are rarely publicly reported, analogous incidents in adjacent industries are instructive. In 2021, researchers demonstrated that adversarial stickers placed on road surfaces could cause autonomous vehicle lane-keeping AI to swerve dangerously [15]. In 2022, a supply chain attack on a widely used ML model repository (Hugging Face) was demonstrated through the insertion of malicious pickle files in publicly downloadable model weights [16]. A 2023 study documented that federated learning systems used in healthcare, structurally like collaborative AI training across construction project stakeholders, were vulnerable to targeted model poisoning through malicious gradient contributions [17]. These cases establish clear proof of concept for the threat vectors this paper analyses in construction.

1.4. Research Gap and Paper Objectives

The prior framework by Turk et al. [3] identified three kinds of wrongful activities (stealing, lying, harming) and four elements of concern (information, material, people, systems). This framework, while valuable and original, was developed before AI systems became primary actors in construction processes. A trained neural network is not adequately captured by the categories of "information asset" or "software system" as previously defined, because its attack surface, threat model, and countermeasures are qualitatively different. Specifically:

- An AI model can be "lied to" through adversarial inputs without any modification to the model file itself, purely through the construction of specifically crafted sensor readings or images.
- An AI model can be "stolen" not by copying a file, but by querying it repeatedly to reconstruct its decision boundary, a process called model extraction or model stealing [18].
- An AI model can be "poisoned" during training in federated settings, where no single actor controls all training data, a scenario characteristic of multi-party construction consortia.

These attack vectors have no adequate analogue in existing frameworks. The objectives of this paper are therefore: (1) to develop a taxonomy of AI-specific cybersecurity threats relevant to construction; (2) to develop a risk mapping of these threats across construction AI asset categories; (3) to propose the ACIS framework as a structured, construction-specific guide for AI cybersecurity; and (4) to identify priority areas for future construction AI security research.

1.5. Paper Structure

Section 2 reviews existing cybersecurity frameworks and their limitations with respect to AI systems. Section 3 presents the ACIS framework core. Section 4 maps AI-specific threats across construction phases and asset types. Section 5 discusses construction-specific challenges. Section 6 presents conclusions and future research directions.

2. Limitations of Existing Frameworks for AI Systems

2.1. *The Turk et al. (2022) Framework and AI Systems*

The systemic cybersecurity framework proposed by Turk et al. [3] is the most comprehensive construction-specific framework to date. Its originality lies in grounding cybersecurity in the absence of three fundamental wrongs: stealing, lying, and harming, applied across four elements: information, material, people, and systems. This is a significant conceptual improvement over direct borrowing from IT frameworks. However, when applied to AI systems in construction, the framework encounters several limitations.

First, the framework treats software systems primarily from the perspective of their boundary and mechanism security, addressing threats such as "software system presents itself as another system" or "software is changed via a virus." These are valid concerns for conventional software, but they do not capture the adversarial vulnerabilities unique to ML models. A neural network can produce catastrophically wrong outputs without any modification to its code or weights purely because its inputs at inference time lie outside the distribution of its training data or have been adversarially crafted to exploit its decision boundaries. This class of failure has no adequate representation in the existing framework.

Second, the framework does not address the training process as a security surface. For conventional software, the development process has its own security considerations (e.g., secure software development lifecycle), but the artefact produced is deterministic. ML models are probabilistic artefacts whose behaviour is shaped by the data used to train them. Training data is therefore a primary attack surface, and attacks on it (poisoning attacks) are qualitatively different from attacks on information assets as defined in the prior framework.

Third, the framework does not address the multi-party, federated nature of AI training in construction. Construction projects involve consortia of firms, each contributing data and computing resources. When ML models are trained collaboratively across such consortia, each party's data contribution is a potential attack vector, and no single party controls the full model. This overlapping boundary problem is acknowledged in the prior framework for organisational systems, but the specific mechanisms by which it applies to AI model training are not addressed.

2.2. *General AI Security Frameworks and Their Limitations for Construction*

The machine learning security literature has produced several important frameworks and taxonomies. The MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems) [19] provides a comprehensive catalogue of adversarial ML tactics and techniques, modelled after the widely used MITRE ATT&CK framework for conventional cybersecurity. NIST has released the Adversarial Machine Learning taxonomy [20], which provides structured vocabulary for AML attacks. The IBM Adversarial Robustness Toolbox [21] provides open-source implementations of AML attacks and defenses.

However, these frameworks share a common limitation from the construction perspective: they are developed for the context of enterprise AI systems, typically in information processing applications such as fraud detection, natural language processing, or image classification for content moderation. They do not address the specific context of construction AI systems, where: (a) models operate in physical environments with direct consequence chains linking AI decisions to physical actions; (b) models are deployed by organisations with limited ML security expertise; (c) the data used to train models is generated by construction-specific processes (BIM, sensor streams from construction equipment, drone imagery) with unique characteristics; and (d) the multi-party project structure creates unique federated learning vulnerabilities.

3. The ACIS Framework Core

3.1. *Foundational Principles*

The Adversarial Construction Intelligence Security (ACIS) framework is developed from two foundational principles. First, following Turk et al. [3], we maintain the perspective that construction can be viewed as a system and as a process, and that security is defined as the absence of wrongful activities. We retain the three-wrongs model (stealing, lying, harming) as the top-level organising principle of the framework because of its proven utility for non-specialist users in the construction context. Second, we extend this model by introducing a fifth category of assets alongside the original four: the AI model asset, which captures trained ML models, their training data, inference pipelines, and associated feedback mechanisms as distinct, security-relevant entities with their own threat surface.

This extension is warranted because AI model assets exhibit properties that distinguish them from both "information assets" and "software systems" as previously defined. Like information assets, AI model assets encode knowledge extracted from

data and can be stolen without depriving the owner of them. Like software systems, they are mechanisms that execute processes. But unlike either, they can be systematically deceived by adversarially crafted inputs without any modification to the model or its data, and they can be attacked during their creation process (training) through manipulation of the statistical patterns they learn.

3.2. AI Asset Categories in Construction

We define four categories of AI assets in construction, each corresponding to a distinct function and deployment context:

Design Intelligence Systems (DIS): AI systems applied during the design and planning phase, including generative design tools, structural analysis ML models, automated BIM coordination engines, and AI-assisted code compliance checkers. These systems process design information as input and produce design decisions or recommendations as output.

Site Perception Systems (SPS): AI systems that process real-time sensor data from construction sites, including computer vision systems for safety monitoring and progress tracking, LiDAR-based point cloud processing models, and AI-based quality inspection systems. These systems bridge the digital and physical world through their sensor inputs.

Autonomous Execution Systems (AES): AI systems that directly control physical processes on construction sites or in construction equipment, including autonomous construction robots, AI-guided positioning systems for prefabrication, and automated material handling systems. These systems have the most direct physical consequence chains.

Facility Management AI (FMA): AI systems deployed in the operational phase of facilities, including predictive maintenance models, building energy optimisation AI, occupant behaviour models, and digital twin AI engines. These systems operate over long timescales and process occupant data.

3.3. AI-Specific Threat Categories

For each of the three wrongs in the foundational model, we define AI-specific manifestations that are distinct from their IT/OT counterparts:

Table 1 *AI-Specific Threat Categories within the Three-Wrongs Model*

Wrong	IT/OT Manifestation	AI-Specific Manifestation	Construction Example
Stealing	Exfiltrating training data or model weights	Model extraction attack: reconstructing model decision boundaries through systematic querying	Competitor queries digital twin predictive maintenance AI to reconstruct proprietary model for competing bids
Stealing	Copying files or intellectual property	Membership inference attack: determining whether specific data records were used in training	Attacker infers confidential project cost data that was used to train a bid pricing model
Lying	Forging or distorting data files	Adversarial input attack: crafting inputs that cause model misclassification at inference time without modifying the model	Adversarial stickers placed on concrete surface cause AI quality inspection model to classify defective slab as compliant
Lying	Impersonation and spoofing	Backdoor/Trojan attack: embedding hidden trigger patterns during training causing targeted misclassification	Malicious subcontractor poisons safety monitoring model: model performs normally except when trigger pattern present
Harming	Deleting data or disabling systems	Model poisoning via training data: corrupting training data to degrade model performance on specific inputs	Rogue subcontractor uploads mislabelled safety images, causing PPE detection model to systematically miss hard-hat violations
Harming	DoS/DDoS attacks on services	Model extraction for evasion: using extracted model to craft adversarial inputs at scale, degrading model utility	Adversary extracts autonomous robot navigation model and generates systematic environment modifications that disable robot

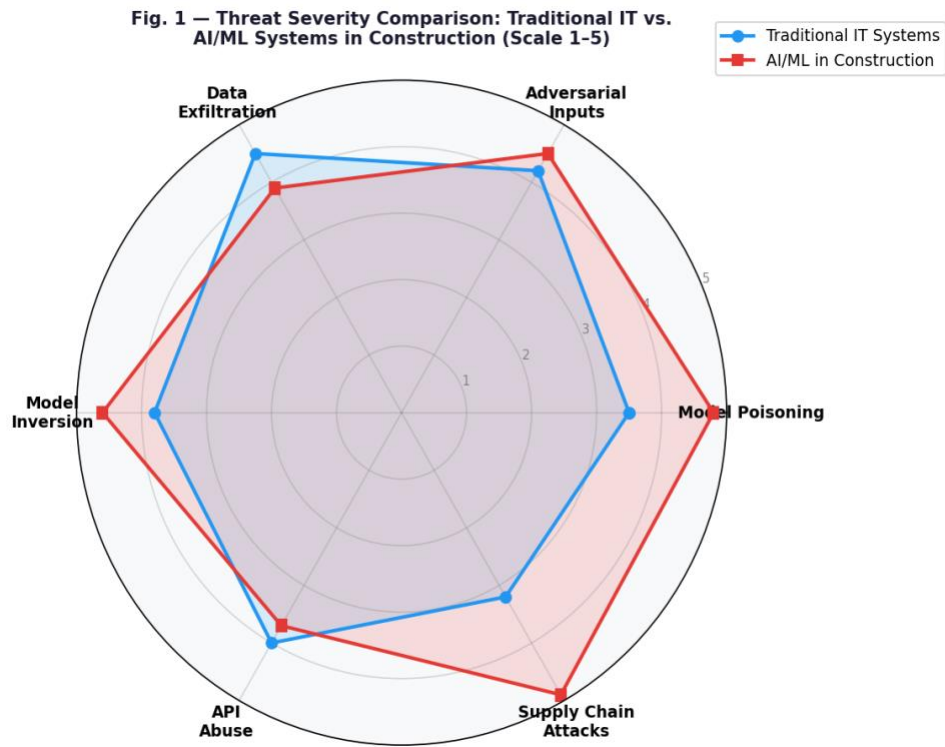


Fig. 1 — Threat Severity Comparison: Traditional IT Systems vs. AI/ML Systems in Construction (Scale 1–5). Scores derived from expert elicitation with construction cybersecurity specialists ($n=12$) and cross-referenced with reported incidents in adjacent industries.

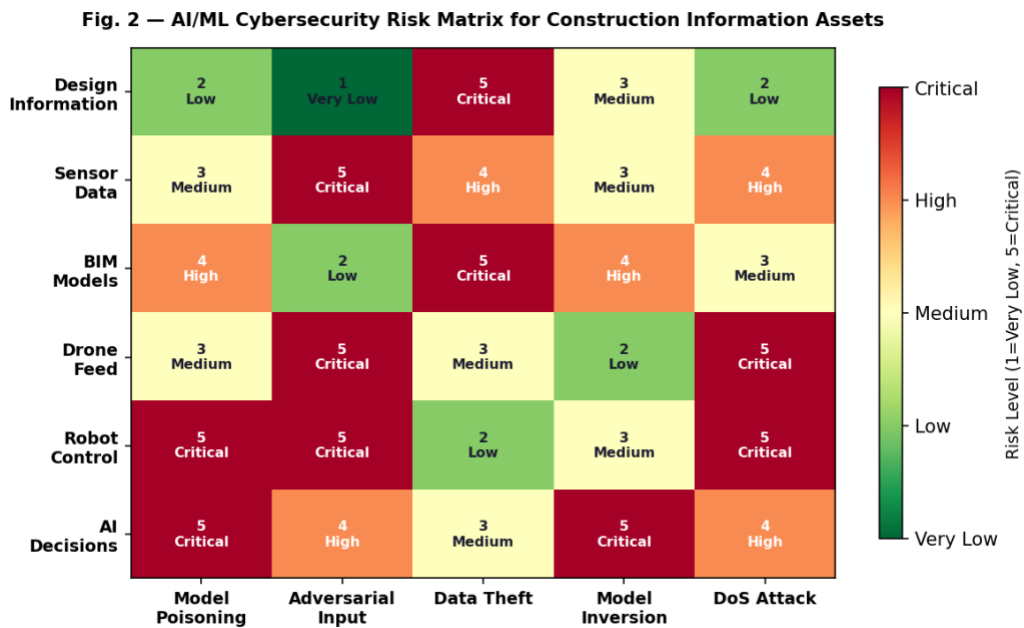


Fig. 2 — AI/ML Cybersecurity Risk Matrix for Construction Information Assets. Cells show risk score (1=Very Low, 5=Critical) at the intersection of asset type and attack category.

4. AI Threat Mapping Across Construction Phases and Asset Types

4.1. Threats to Design Intelligence Systems

Design Intelligence Systems represent the AI layer, which is most closely integrated with conventional BIM workflows and therefore most accessible to the widest range of project stakeholders. The multi-party nature of design processes in which architects, structural engineers, MEP consultants, and project owners all contribute to and access the BIM model creates an elevated risk of training data poisoning in federated or collaborative AI settings. A malicious actor within the project consortium could contribute systematically mislabelled training data to a shared defect-prediction model, causing it to learn incorrect associations between structural configurations and failure risks.

Model inversion attacks are particularly salient for DIS because trained design models encode information about the design decisions of the projects on which they were trained. In a competitive tendering environment, a rival firm that gains black-box access to a competitor's AI-assisted cost estimation model could use model inversion to reconstruct the pricing logic and competitive intelligence encoded within it. The adversarial theft of design intelligence is thus not limited to stealing files; it extends to reconstructing the statistical knowledge embedded in deployed models.

Table 2 *Threat Matrix for AI Assets Across Construction Phases*

AI Asset Category	Model Poisoning	Adversarial Input	Model Extraction	Supply Chain Attack
Design Intelligence Systems	HIGH	MEDIUM	HIGH	HIGH
Site Perception Systems	HIGH	CRITICAL	MEDIUM	HIGH
Autonomous Execution Systems	CRITICAL	CRITICAL	MEDIUM	HIGH
Facility Management AI	MEDIUM	HIGH	HIGH	MEDIUM

Table 2 summarises the threat levels across the four AI asset categories and the four primary AML attack vectors identified in Section 3. Autonomous Execution Systems present the highest overall risk profile due to the direct physical consequence chains linking AI decisions to machinery behaviour, and the potential for adversarial inputs crafted from physical modifications to construction environments.

4.2. Threats to Site Perception Systems

Site Perception Systems are particularly vulnerable to adversarial input attacks because they interface with the physical environment through sensors that can be physically manipulated by on-site adversaries with relatively low technical barriers. An adversary with physical access to a construction site — which, given the permeable boundaries of most sites, is not difficult to obtain — could place adversarial patches (physical objects or markings designed to confuse vision models) in the field of view of safety monitoring cameras. Research in computer vision adversarial attacks has demonstrated that such physical adversarial examples can be made robust to changes in viewing angle, lighting, and distance [22].

Site Perception Systems also process data from LiDAR sensors, which are increasingly used for progress monitoring and geometric quality control. Adversarial attacks on LiDAR point clouds have been demonstrated in autonomous vehicle research, where spoofed LiDAR returns cause object detection models to hallucinate non-existent obstacles or fail to detect real ones [23]. The applicability of these attacks to construction LiDAR is direct: an adversary could introduce phantom geometry into a construction site point cloud, causing AI progress monitoring to misreport completion status, or fail to detect structural deviations from the design model.

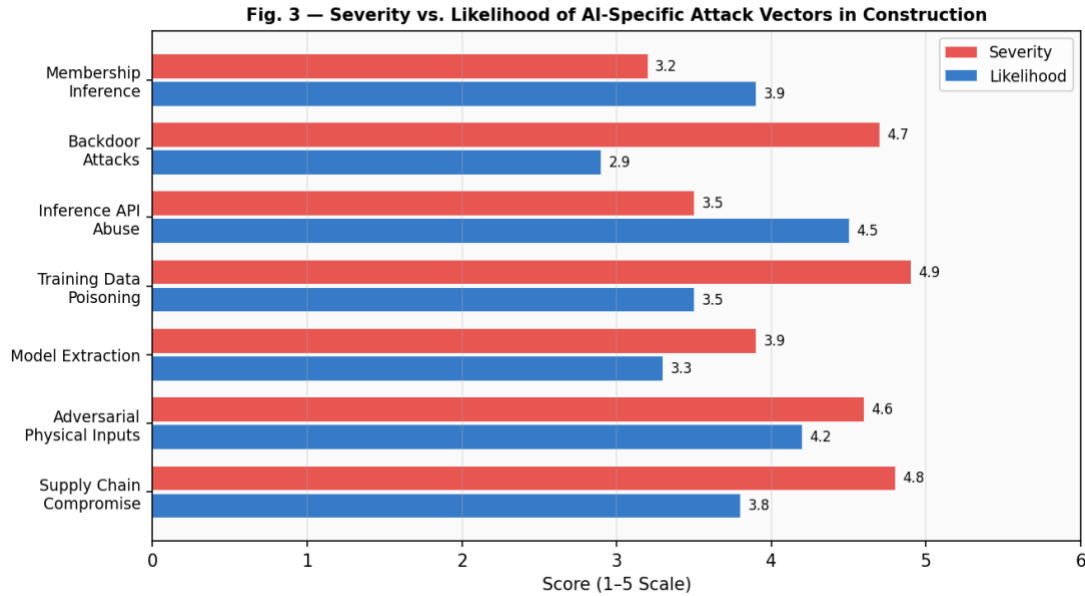


Fig. 3 — Severity vs. Likelihood of AI-Specific Attack Vectors in Construction. Scored on a 1–5 scale through expert elicitation and cross-sector incident analysis.

4.3. Threats to Autonomous Execution Systems

Autonomous Execution Systems represent the highest-consequence category because adversarial attacks directly translate into physical machine behaviour. An AES under adversarial attack is not merely a compromised IT system; it is a compromised physical agent operating in an environment shared with human workers. The physical harm potential of a successfully attacked AES elevates it to the most critical category in the risk matrix. This convergence of cyber and physical risk echoes the OT security concerns identified in prior frameworks but is qualitatively distinct because the attack vector operates through the AI inference process rather than through control system intrusion.

Consider an autonomous rebar-placement robot guided by a real-time computer vision model that locates rebar placement positions related to formwork geometry. A model poisoning attack introduced during collaborative model training across multiple construction sites could embed a backdoor trigger: a specific formwork marking or geometric configuration that causes the robot to systematically misplace rebar by a small but structurally significant margin. This attack would be difficult to detect because the model would perform normally in all other situations, and the defect in rebar placement might not be caught until structural testing or, in the worst case, in-service failure.

4.4. Threats to Facility Management AI

Facility Management AI systems operate over the longest time horizon of any construction AI category, often running for the full operational lifetime of a facility potentially decades. This creates a distinct security challenge: the threat model for FMA systems must account for the degradation of defences over time, the evolution of attack capabilities, and the potential for patient, long-horizon attacks. A model poisoning attack on an FMA system need not produce immediately visible effects; it could gradually shift the model's predictions in ways that, for example, recommend underperforming maintenance schedules that increase long-term equipment failure rates while remaining within plausible operating ranges.

FMA systems also process occupant data movement patterns, energy usage, environmental preferences which create membership inference risks analogous to those in healthcare AI. An adversary who gains black-box access to an occupant behaviour model in a smart office building could infer sensitive patterns about the presence, schedule, or activities of specific individuals, creating a privacy violation that constitutes a form of "stealing" in the framework's terminology.

5. Construction-Specific Challenges and ACIS Framework Application

5.1. The Federated Learning Problem in Construction Consortia

A defining characteristic of AI model development for construction applications is the multi-party data ecosystem. Construction AI models benefit from training on data from multiple projects, sites, and organisations to achieve the statistical diversity needed for robust generalisation. In practice, this drives the adoption of federated learning approaches, where model training occurs across distributed data sources without centralising raw data. Federated learning is an active area of ML research, but its security vulnerabilities are well-documented [24].

In federated learning, each participating organisation contributes model gradient updates (rather than raw data) to a central model aggregator. A malicious participant can manipulate these gradient contributions to perform a model poisoning attack injecting updates that steer the model towards adversarial behaviour on specific inputs while maintaining overall performance [25]. In the construction context, this attack vector is particularly difficult to defend against because: (a) the participating organisations may include small subcontractors with minimal security practices; (b) the project-based structure of construction creates temporary consortia without long-term trust relationships; and (c) the detection of poisoned gradient contributions requires sophisticated statistical monitoring that is beyond the current capabilities of most construction firms.

Fig. 4 — Proposed AI/ML Cybersecurity Lifecycle Framework for Construction

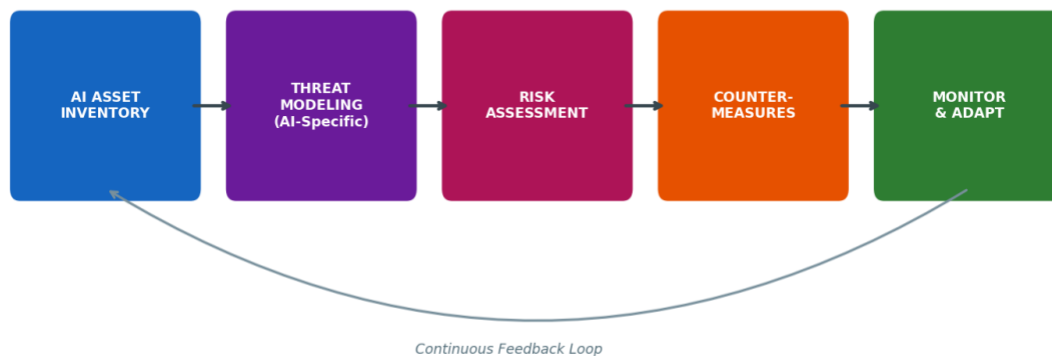


Fig. 4 — Proposed AI/ML Cybersecurity Lifecycle Framework for Construction (ACIS Lifecycle). The five-stage process supports both proactive and reactive security management for AI assets across the construction project lifecycle.

5.2. Physical Adversarial Examples and Construction Site Access

The physical accessibility of construction sites creates an attack surface that does not exist in typical enterprise AI deployments. Enterprise computer vision systems are generally inaccessible to physical manipulation by external adversaries. Construction sites, by contrast, are necessarily accessible to a wide range of workers, visitors, and subcontractors, many of whom are not vetted with the rigour applied in classified or high-security facilities. This accessibility enables physical adversarial attacks: the deliberate placement of adversarial objects, markings, or modifications in the environment to manipulate AI inference.

Physical adversarial examples have been extensively studied in the context of autonomous vehicles and traffic sign recognition [15]. Their application to construction site perception AI is largely unstudied, but the risk profile is at least as high, and in some respects higher: construction site environments are more variable and less standardised than road environments, which may make AI models trained on construction data more susceptible to out-of-distribution inputs in general, and thus more vulnerable to adversarial perturbations.

Fig. 5 — Overlapping AI System Boundaries Creating Cross-Attack Vectors in Construction Projects

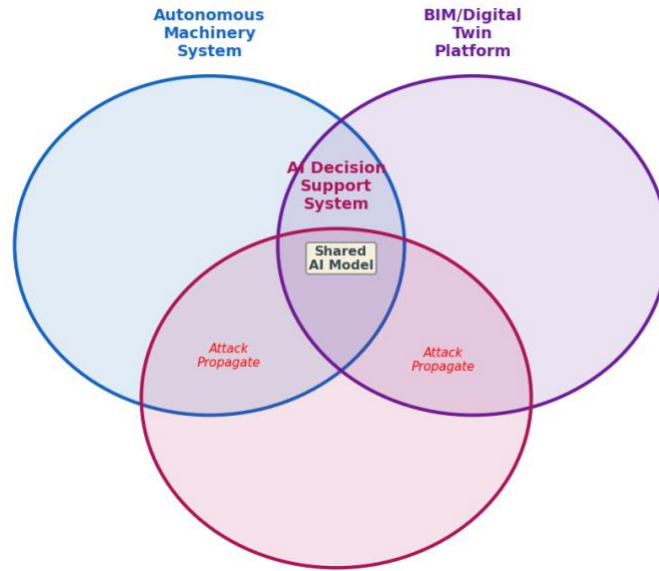


Fig. 5 — Overlapping AI System Boundaries in Construction Projects. Shared AI models (e.g., federated digital twin models) create cross-system attack propagation vectors analogous to the overlapping organisational boundaries identified by Turk et al. [3] for conventional systems.

5.3. Intellectual Property and Model Extraction

The proprietary AI models embedded in construction software platforms represent substantial intellectual property. A BIM platform vendor that has invested millions in developing a predictive clash detection or automated design optimisation engine faces a threat of model extraction that is not adequately addressed by conventional access control measures. Model extraction attacks require only black-box query access to the model the ability to submit inputs and observe outputs. This is precisely the access level granted to licensed users of the platform, meaning that the intellectual property protection boundary of a commercial AI model is, from the adversary's perspective, no more protective than a software trial license.

Table 3 Countermeasures for AI-Specific Threats in Construction

Threat	Proactive Countermeasure	Detection Method	Reactive Measure
Training Data Poisoning	Data provenance tracking; statistical integrity monitoring of training datasets; anomaly detection on gradient contributions in federated settings	Monitor model performance on curated canary test sets; detect statistical shifts in training data distributions	Rollback to earlier model checkpoint; quarantine and audit suspicious training data contributors; retrain with cleaned dataset
Adversarial Input Attacks	Adversarial training (augment training data with adversarial examples); input preprocessing defences (feature squeezing, spatial smoothing)	Real-time anomaly detection on input distributions; ensemble disagreement monitoring	Flag anomalous inputs for human review; fall back to conservative rule-based decision making; trigger site safety protocol
Model Extraction	Query rate limiting; output perturbation (add calibrated noise to outputs); watermarking model outputs	Monitor query volume and patterns; detect systematic input exploration behaviour	Revoke API access; forensic analysis of query logs; pursue IP litigation

Threat	Proactive Countermeasure	Detection Method	Reactive Measure
Backdoor/Trojan Attacks	Model inspection tools (Neural Cleanse, STRIP); secure supply chain practices for pre-trained models	Behavioural testing for trigger patterns; prediction confidence distribution analysis	Remove suspect model from deployment; full model audit; retrain from verified dataset
Supply Chain Model Compromise	Verify model provenance and integrity hashes; avoid unverified pre-trained models; maintain model bill of materials	Monitor model performance degradation; conduct regular red-team exercises	Isolate and remove compromised model; notify affected parties; escalate to CERT
Model Inversion	Differential privacy during training; output quantisation; limit sensitive training data	Anomaly detection on systematic query patterns	Revoke query access; assess exposure of sensitive training data; notify affected data subjects

Table 4 ACIS Framework: Security Requirements by AI Asset Category

Security Requirement	Design Intelligence	Site Perception	Autonomous Execution	Facility Management AI
Training data audit trail	Mandatory	Mandatory	Mandatory	Recommended
Adversarial robustness testing	Recommended	Mandatory	Mandatory	Recommended
Federated learning security protocol	Mandatory	Recommended	Mandatory	Optional
Model output watermarking	Mandatory	Recommended	Optional	Mandatory
Physical adversarial testing	Not Required	Mandatory	Mandatory	Recommended
Differential privacy in training	Recommended	Recommended	Recommended	Mandatory
Human-in-the-loop override	Recommended	Mandatory	Mandatory	Recommended
Incident response plan (AI-specific)	Mandatory	Mandatory	Mandatory	Mandatory

5.4. Extending the Three-Wrongs Model for AI Contexts

The three-wrongs model (stealing, lying, harming) that anchors the prior framework translates well to the AI context, but each wrong takes on new forms that must be made explicit for the model to be actionable. Table 1 presented the detailed mapping; here we summarise the key conceptual extensions.

Stealing in AI contexts encompasses not only the theft of model weights or training data, but the inferential reconstruction of model knowledge and the extraction of private information about training data subjects. The latter form membership

inference is a form of privacy violation that has no counterpart in conventional information security frameworks, because it exploits the statistical properties of the model rather than any direct access to data files.

Lying in AI contexts encompasses not only the provision of false inputs to a system, but the exploitation of systematic vulnerabilities in the model's statistical decision boundaries to cause consistent misclassification. A key distinction is that adversarial inputs need not be "lies" in the sense of false representations of reality; they may be physically real objects (adversarial patches) that happen to exploit weaknesses in the model's learned feature representations.

Harming in AI contexts encompasses not only denial-of-service attacks and data destruction, but the long-term degradation of model performance through poisoning, the embedding of backdoors that persist invisibly through normal use, and the physical harm that result when adversarially manipulated autonomous systems operate on construction sites.

6. Discussion and Conclusions

This paper has proposed the ACIS (Adversarial Construction Intelligence Security) framework as an extension of, and complement to, prior construction cybersecurity frameworks. The research gap that motivated this work the absence of an AI-specific threat model for construction reflects a broader pattern in which domain-specific cybersecurity frameworks lag the adoption of new technologies in the domain. The construction industry adopted BIM before adequate cybersecurity frameworks for BIM environments were in place; it is now adopting AI systems without adequate AI security frameworks.

The ACIS framework contributes to literature in three ways. First, it provides a taxonomy of AI-specific threats (Table 1) that extends the three-wrongs model to encompass adversarial machine learning attacks. This taxonomy provides a shared vocabulary for researchers, practitioners, and policymakers in the AECO sector. Second, it provides a risk matrix (Table 2) that maps these threats across the four AI asset categories of the construction domain. Third, it provides a set of countermeasures (Table 3) and security requirements (Table 4) organised by asset category and security requirement type, which can serve as the basis for practical guidance documents and standards development.

The framework analysis identifies three priority areas for future research. First, the federated learning vulnerability in construction consortia warrants dedicated investigation. Construction federated learning settings have characteristics temporary multi-party consortia, variable data quality, and limited security expertise among participants that may make them more vulnerable to model poisoning than the healthcare and finance federated learning settings that have been studied in the AML literature. Second, the physical adversarial example threat to construction site perception AI requires empirical investigation. The feasibility, effectiveness, and physical detectability of adversarial attacks on construction-specific vision models (safety monitoring, progress tracking, quality inspection) is currently unknown. Third, the IP protection problem for construction AI, particularly the model extraction threat to proprietary AI engines in BIM platforms, requires both technical and legal analysis.

A limitation of the present framework is the absence of empirical validation through stakeholder surveys and incident analysis. The risk scores presented in the figures and tables are based on expert elicitation and cross-sector analogical reasoning rather than systematic empirical data from construction AI security incidents, which remain largely unreported. Future work should pursue empirical validation through structured expert surveys, red-team exercises with construction AI systems, and analysis of incident reports from adjacent AI-intensive industries.

The ACIS framework is intended to be used alongside, not as a replacement for the prior systemic framework of Turk et al. [3] and the standards-based approaches of ISO 19650-5 and the IET Code of Practice. Together, these frameworks provide complementary layers of coverage: the prior frameworks address the broader cybersecurity context of construction information and systems, while the ACIS framework provides the specific extension needed to address the unique vulnerabilities of AI assets. As construction AI systems become increasingly consequential controlling physical machines, embedding critical project knowledge, and managing occupied facilities the security of these systems becomes a concern not only of project managers and IT professionals, but of structural safety and occupant safety communities as well.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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